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UNIT 7

CHEMICAL KINETICS



Svante August Arrhenius (1859 –1927)

Svante August Arrhenius was a Swedish scientist. Arrhenius was one of the founders of the science of physical chemistry. He focused his attention on the conductivities of electrolytes. He proposed that crystalline salts dissociate into paired charged ions when dissolved in water, for which he received the Nobel Prize for Chemistry in 1903. He also proposed definitions for acids and bases. He formulated the concept of activation energy.

WJN386

Learning Objectives

After studying this unit, the students will be able to

- define the rate and order of a reaction,
- derive the integrated rate equations for zero and first order reactions,
- describe the half life period,
- describe the collision theory,
- discuss the temperature dependence of the rate of a reaction, and
- explain various factors which affect the rate of a reaction.

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